

Education

MSc Machine Learning

KTH Royal Institute of Technology

Stockholm, Sweden

Aug 2026 – Present

- Relevant Coursework: Financial Mathematics, Martingales and Stochastic Integrals, Regression Analysis, Reinforcement Learning, Time Series Analysis

MSc Telecommunications Engineering - Academic Focus on AI

Top 5%

Barcelona, Spain

Polytechnic University of Catalonia (UPC)

Sept 2025 – Present

- Relevant Coursework: Computer Vision with Deep Learning, Deep Learning for Artificial Intelligence, Machine Learning from Data
- Honours: Deep Learning for Artificial Intelligence, Electronic Instrumentation and Optoelectronics

BSc Telecommunications Systems Engineering

Top 5%

Barcelona, Spain

Autonomous University of Barcelona (UAB)

Sept 2020 – Jul 2025

- Honours: Algebra, Analogue Electronics, Calculus

BSc Computer Engineering (Specialisation in Information Technology)

Top 5%

Barcelona, Spain

Autonomous University of Barcelona (UAB)

Sept 2020 – Jul 2025

- Honours: Artificial Intelligence, Ethics

Experience

Machine Learning Research Intern (Hybrid)

Barcelona, Spain

Barcelona Supercomputing Center (BSC) - Earth Science, Artificial Intelligence Team (AIT)

Mar 2026 – Aug 2026

Research on Graph Neural Networks (GNNs) for spatiotemporal interpolation and forecasting of Earth system variables

- Designed a Graph Attention Network for time series interpolation and forecasting on ERA5 and ClimateDT datasets, generalized to a hierarchical multi-mesh U-Net architecture.
- Built distributed multi-GPU, multi-node training pipelines with PyTorch DDP and SLURM.
- Investigated de-cycling of the training signal via Fourier regression on seasonal and diurnal components.

RF and Microwave Research Assistant (Hybrid)

Barcelona, Spain

Universitat Oberta de Catalunya (UOC) - Wireless Networks Research Group (WiNe)

Jul 2024 – Jun 2025

Research on antenna-based sensors, electromagnetic characterization of materials and passive RFID systems

- Designed and built antennas for RF and microwave applications, used to develop passive RFID-based wireless sensors.
- Performed electromagnetic simulations and characterization of radiating systems and RF components.
- Programmed control and reading functions for an RFID reader module in C, C#, and Java, integrating a client-server architecture based on Node.js and Vue running on a Raspberry Pi.

Publications

Journal papers

Passive Monitoring of Water Evaporation in Textile Fabrics for Industrial Applications

Aug 2025

Using a Battery-Free UHF RFID Tag (*IEEE Journal of Radio Frequency Identification*)

P Brunet-Gil, S López-Soriano

[10.1109/JRFID.2025.3600837](https://doi.org/10.1109/JRFID.2025.3600837) [🔗](#)

Remote Identification of Liquids Using Absorbent Materials: a Passive UHF RFID-Based

Jan 2025

Method (*IEEE Sensors Journal*)

S López-Soriano, *P Brunet-Gil*, M Alsultan, J Meliá-Seguí

[10.1109/JSEN.2024.3524909](https://doi.org/10.1109/JSEN.2024.3524909) [🔗](#)

Conference papers

Experimental Analysis of UHF RFID Tags Mounted on PET Liquid Containers Subjected to External Forces (19th European Conference on Antennas and Propagation (EuCAP))

Apr 2025

P Brunet-Gil, M Alsultan, J Meliá-Seguí, S López-Soriano

[10.23919/EuCAP63536.2025.10999483](https://doi.org/10.23919/EuCAP63536.2025.10999483) [🔗](#)

A Method for the Dielectric Characterization of Liquids Combining Power Laws of Absorbent-Liquid Mixtures (19th European Conference on Antennas and Propagation (EuCAP))

Apr 2025

S López-Soriano, *P Brunet-Gil*, M Alsultan, J Meliá-Seguí

[10.23919/EuCAP63536.2025.10999864](https://doi.org/10.23919/EuCAP63536.2025.10999864) [🔗](#)

Certifications

Elements of AI

Jan 2024

University of Helsinki

certificates.mooc.fi/validate/gi6wfgq3ncs [🔗](#)